

CamSketch: Authoring Camera Trajectories for 3D Virtual Concerts via Example-Derived Codes and User Sketching

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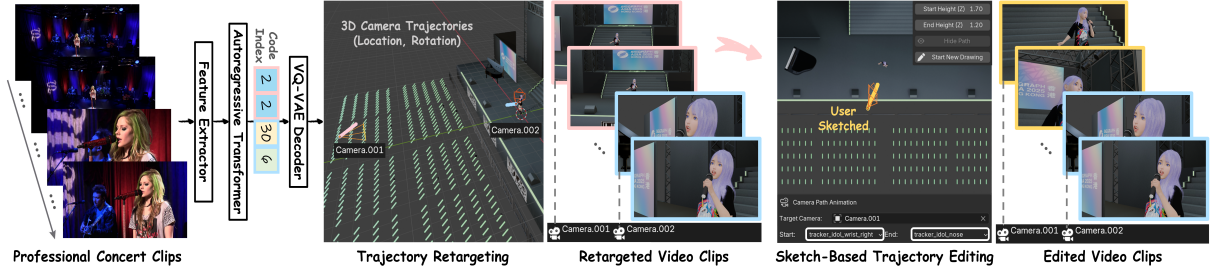


Figure 1: CamSketch predicts and retargets structured camera motion codes from professional concert footage to virtual concert environments and provides an intuitive scene-aligned bird’s-eye view interface for sketch-based trajectory editing.

Abstract

We present CamSketch, a user-centric, example-driven camera motion controller for authoring immersive 3D trajectories in virtual concerts. CamSketch learns structured camera motion codes from professional concert footage and enables users to freely sketch and refine camera paths through an intuitive, scene-aligned interface. Built on a vector quantised-variational autoencoder (VQ-VAE) and an autoregressive transformer, it learns discrete camera motion codes and predicts code index sequences directly from raw footage. The predicted codes are retargeted to virtual concerts and remain editable through user-sketched trajectories on a spatial canvas. Results show that CamSketch bridges example-driven automation and user-guided creativity, enabling high-quality camera motions for virtual concert performances.

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1 Introduction

Controlling camera motion in virtual concerts plays a critical role in crafting immersive and emotionally resonant viewing experiences. Recent advances in example-driven camera control—primarily developed for professional footage such as films—have enabled the

automatic generation of camera trajectories for film-style virtual scenes [Jiang et al. 2021, 2020]. While these approaches reduce manual effort and produce visually appealing results, they remain constrained by specific scene configurations—such as requiring exactly two characters to infer plausible camera motion dynamics. More importantly, current systems offer limited flexibility for users to intuitively adjust or personalize the generated camera trajectories. This lack of a direct and accessible editing mechanism creates a gap between automated generation and creative control. To bridge this gap, we introduce a user-centric framework that not only learns to imitate example concert trajectories, but also enables intuitive refinement through sketch-based trajectory editing.

In this work, we present CamSketch, a user-centric, example-driven camera motion controller tailored for virtual concerts (see Figure 1). CamSketch is designed to empower both automation and user control through three core components: (i) a vector quantised-variational autoencoder (VQ-VAE) (Figure 2 (a)) that discretizes complex 3D camera trajectories—including both locations and rotations—into structured camera motion codes, thereby transforming the regression problem into a more tractable classification task and enabling interpretable camera choreography; (ii) an autoregressive transformer (Figure 2 (b)) that learns to predict these camera motion code indices from raw concert footage, without relying on specific scene configurations, and effectively models the relationship between visual content and camera motion dynamics; and (iii) an interactive retargeting module (Figure 1) that retargets the predicted codes to virtual concert environments while allowing users to freely create and refine camera paths through intuitive sketch-based editing. This integration of learning and interaction closes the loop between example-driven automation and user-guided creativity. We show that CamSketch enables the generation of high-quality, customizable 3D camera motions for virtual concerts, facilitating immersive visual storytelling while preserving user agency.

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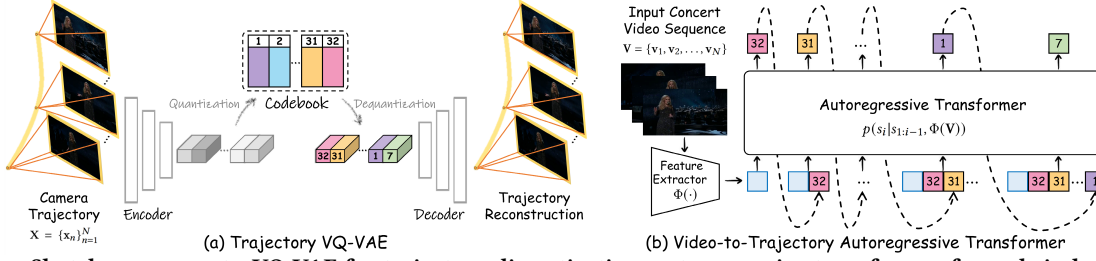


Figure 2: CamSketch components: VQ-VAE for trajectory discretization; autoregressive transformer for code index prediction.

Table 1: Quantitative Evaluation of Camera Trajectories

Method	Location Error (ED) ↓	Rotation Error (AE) ↓
Baseline	1.27m	47.29°
Ours	0.94m	16.26°

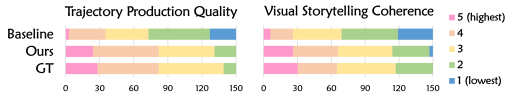


Figure 3: Subjective 5-point opinion scores from 30 raters.

2 Our Approach

Given a camera trajectory $X = \{\mathbf{x}_n\}_{n=1}^N$ with N frames, where $\mathbf{x}_n \in \mathbb{R}^6$ denotes the camera’s location (x, y, z) and Euler rotation $(\theta_x, \theta_y, \theta_z)$ in 3D Cartesian coordinates, we first train a VQ-VAE—comprising an encoder, a learnable codebook, and a decoder—to discretize the trajectory into structured camera motion codes, jointly optimized for trajectory reconstruction (see Figure 2 (a)). With the trained VQ-VAE, the camera trajectory X can be represented as a sequence of motion code indices $S = \{s_1, s_2, \dots, s_{N/\omega}\}$, where each $s_i \in \{1, 2, \dots, K\}$ is an index from the learned codebook ($K = 32$), and $\omega = 16$ represents the temporal downsampling rate, encoding every consecutive 16 frames into a single motion code index. Accordingly, we formulate the example-driven camera motion control as an autoregressive next-index prediction task solved by an autoregressive transformer (see Figure 2 (b)). Given previous motion code indices $s_{1:i-1}$ and an exemplar raw concert video sequence $V = \{v_1, v_2, \dots, v_N\}$ —where each frame $v_n \in \mathbb{R}^{H \times W \times C}$ denotes an RGB image with height H , width W , and C channels—the autoregressive transformer predicts the probability distribution of the next camera motion code index as $p(s_i | s_{1:i-1}, \Phi(V))$, where $\Phi(\cdot)$ is a feature extractor. The camera motion code index with the highest probability is then selected as the prediction result. This autoregressive prediction process is repeated iteratively until the length of the predicted motion code index sequence matches $\lfloor \frac{N}{\omega} \rfloor$, corresponding to the temporally downsampled length of the input raw concert video sequence V . After predicting the motion code index sequence, we retrieve the corresponding motion codes from the learned codebook and decode them into the camera trajectory, which is then retargeted to the 3D virtual concert environment.

To support intuitive sketch-based editing, we design a spatial canvas that provides a bird’s-eye view of the virtual concert environment (see Figure 1). This canvas aligns with the X-Y plane of the 3D scene, allowing users to sketch 2D trajectories corresponding to the camera’s motion in the X and Y axes. Users can draw camera paths anywhere in the scene and specify the intended subject—such as a

singer (e.g., specific body parts), an instrument, or the audience—for the camera to focus on. Z-axis values are specified separately by the user, enabling complete 3D camera path construction. Maintaining this coordinate mapping allows users to flexibly modify or replace predicted camera trajectories through simple sketch input.

3 Evaluation and Discussion

For VQ-VAE and autoregressive transformer training, we annotate 3D camera location and rotation at the frame level on our self-built concert stage, based on 60 official concert videos (~ 4.5 hours and 450K images), to emulate real-world camera choreography. The dataset covers 51 concerts, 34 singers, and 11 genres, all aligned with singing performances. We split the data into 90% for training and 10% for testing across all experiments. We train the VQ-VAE with reconstruction, embedding, and commitment losses [Oord et al. 2017], and use cross-entropy loss (i.e., classification loss) to train the autoregressive transformer [Chen et al. 2020]. We adopt Euclidean distance (ED) and angular errors (1) to evaluate the accuracy of estimated camera locations and rotations during testing.

$$\text{Angular Errors (AE)} = 2 \cdot \cos^{-1}(|\langle \hat{\mathbf{q}}, \mathbf{q} \rangle|), \quad (1)$$

where $\hat{\mathbf{q}}$ and $\mathbf{q}$ are the predicted and ground-truth camera rotations, both represented as quaternions converted from Euler angles.

We quantitatively compare our approach (VQ-VAE + autoregressive transformer), which formulates example-driven camera control as a classification task, with a regression baseline that directly regresses camera locations and rotations using an autoregressive transformer. As demonstrated in Table 1, our classification-based method outperforms the regression baseline, largely due to the use of VQ-VAE, which simplifies the task by discretizing trajectories into structured camera motion codes that serve as high-level motion primitives for classification. Subjective evaluation with 5-point opinion scores across five sets further indicates that our method produces production-quality trajectories with coherent visual storytelling when retargeted to the virtual concert—outperforming the regression baseline and approaching ground-truth quality (see Figure 3). Regarding sketch-based editing, the mean opinion score of 4.55 reflects high user satisfaction with the bird’s-eye-view spatial canvas used to design and edit camera trajectories. Despite encouraging results, our current approach lacks music conditioning in trajectory generation; we plan to develop music-driven techniques to enhance camera authoring for future virtual concerts.

Acknowledgments

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